Semih Eski

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Skills

Mechanical: SolidWorks, NX, Ansys (FEA), GD&T, Tolerance Stack-up, DFMA, FMEA/Root-Cause Analysis
Electrical / SW: PCBA Design, Altium, Raspberry Pi, ESP32, Instrumentation, Actuators, Soldering, DMM, Oscilloscope, Python, MATLAB, C++
Manufacturing: CNC, Injection Molding, Composite Fabrication, Additive Manufacturing, Lathe, Mill

Experience

Mechatronics Engineering Intern — May 2026 – Present
Electrans Technologies Ltd. — *Oakville, ON*

- Designed & built an automated electro-pneumatic press using SolidWorks to insert 600,000 compression limiters into production parts, cutting cycle time 83%, eliminating operator injury and part-damage risks; produced GD&T drawings for fabrication
- Designed a universal, lightweight, self-centering portable QC jig with a custom sliding two-axis XY stage, verifying installation hole positions to 0.2 mm accuracy across 3,000+ imperfect trailers spanning 30"/36" kingpin setbacks
- Designed & built a custom spring-loaded retractable pin system with counterbore datum locating for insertion alignment, designed for high-cycle reliability through PFMEA, worst-case tolerance stack-up, and fatigue-life sizing of components
- Designed & built a real-time IP67 electromechanical truck-boom-angle measuring system using a hall-effect rotary sensor driven by a four-bar linkage; validated sensor accuracy and repeatability to 0.01° via test setup

Co-Founder & Mechanical Team Lead — Dec 2025 – Present
Waterloo Automation Collective — *Waterloo, ON*

- Co-founded a 10+ member machine design student team; directing the full mechanical design & launch of a motorized autonomous 15-ton injection molding machine with 100 MPa injection capability
- Owning the design & build of a 15-ton five-point toggle clamping mechanism; synthesized linkage kinematics in MATLAB, validated via FEA in ANSYS; releasing GD&T drawings backed by a 50+ variable worst-case 3D stack-up
- Designed and built an automatic ejection unit with a 10×10 pin grid, accommodating manufacturing tolerances, varying mold sizes and geometries, while maintaining constant stroke and clamping tonnage without retooling

Mechanical Project Manager & Designer — Aug 2025 – Aug 2026
Waterloo Aerial Robotics Group (WARG) — *Waterloo, ON*

- Led clean-sheet design of an STOL UAV empennage in SolidWorks, driven by configuration trade studies; root-caused structural failures and automated design for stability-and-control through a MATLAB toolchain
- Engineered a foldable drone-arm mount in SolidWorks, achieving 80% mass reduction; verified durability via a 300-cycle fatigue test
- Manufactured R&D composite airframes via split-mold tooling and multi-method layups to maximize strength-to-weight ratio

Mechanical Engineering Intern — Aug 2024 – Dec 2024
Equans Services — *Montreal, QC*

- Performed complex WC & RSS tolerance stack-up analyses to verify conveyor belt alignment within ±5 mm, across a 6 m long conveyor
- Performed 30+ quality control inspections on production-line equipment, diagnosing & replacing 6+ faulty parts, restoring operation

Projects

Amphibious Drone

- Designed an ROV using SolidWorks Surfacing and parametric master modeling of a 15+ part assembly, maintaining G2 curvature
- Achieved IPX7-level frame with 15% neutral buoyancy margin; instrumented the ROV and acquired data during submersion tests

CubeSat Structural FEA & Vibration Analysis

- Performed non-linear quasi-static FEA on UW Orbital's CubeSat in ANSYS to simulate launch conditions, validating structural integrity
- Cut down computation time by 14 hours, by achieving 81% high-quality hex-dominant mesh via strategic partitioning of the geometry
- Conducted modal and random vibration (PSD G) FEA on the pretensioned chassis by solving for modes until >90% effective mass in each axis, finding low-amplification IMU mounting zones by mapping 3-sigma random-vibration displacement/stress

Autonomous Object-Picking Vehicle

- Achieved 95% real-time detection, ±5 cm distance precision, implementing machine vision on Raspberry Pi using a custom Python/OpenCV vision pipeline for detecting ping pong balls and range estimation; debugging real-time data via iterative tests
- Developed C++ PID navigation using IR & ToF sensors and a low-latency UART between RPi & ESP32 to execute real-time commands

Education

University of Waterloo – GPA 3.9/4.0 — Aug 2025 – Present
Candidate for BASc - Honours Mechanical Engineering, Co-op — *Waterloo, ON*

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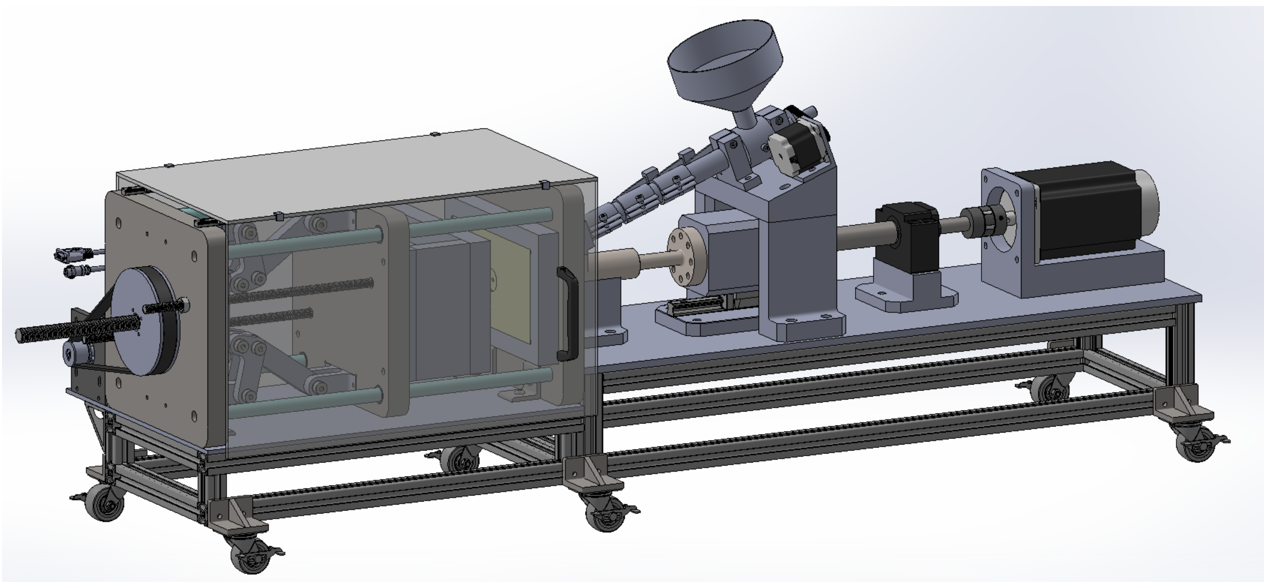
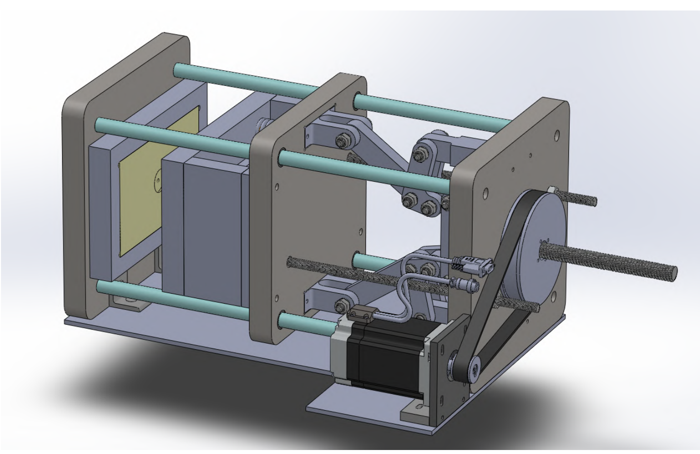
Motorized V-Line Injection Molding Machine @ WAC

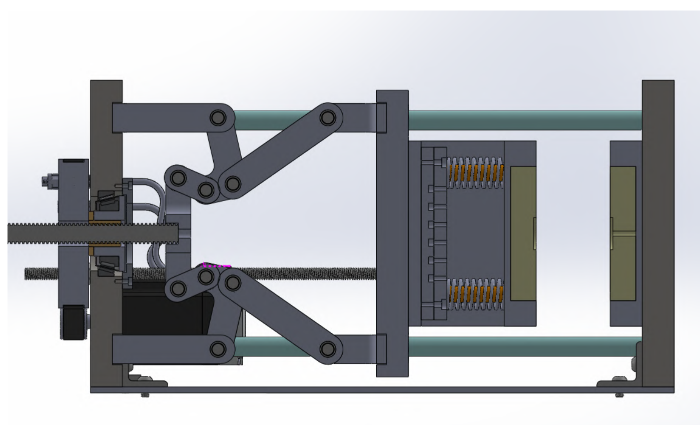
- Co-founded a 10+ member machine design student team & leading the mechanical design of a motorized V-line injection molding machine (15-ton clamping, 100MPa injection)
- Sized screws, bearings, and structure against buckling, thread stripping, yielding, fracture, fatigue, thermal expansion, and brinelling; designed guides against binding and stick-slip, and isolated components from cycle vibration
- Synthesized toggle linkage kinematics in MATLAB to meet tonnage, stroke, and mechanical-advantage targets
- Producing a 50+ variable 3D worst-case tolerance stack-up and GD&T drawings for fabrication



Injection Molder Full CAD



Clamping, Ejection, and Mold (CEM) Units View



CEM Section View

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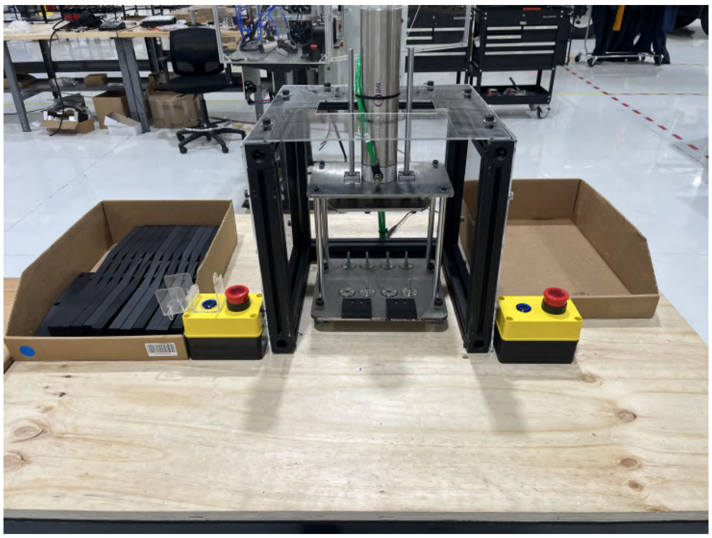
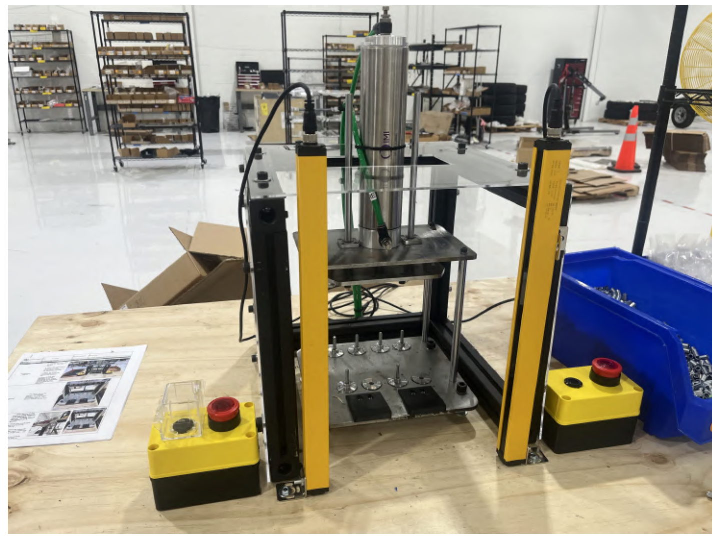
Automated Electro-pneumatic Precision Insertion Machine @ Electrans Technologies

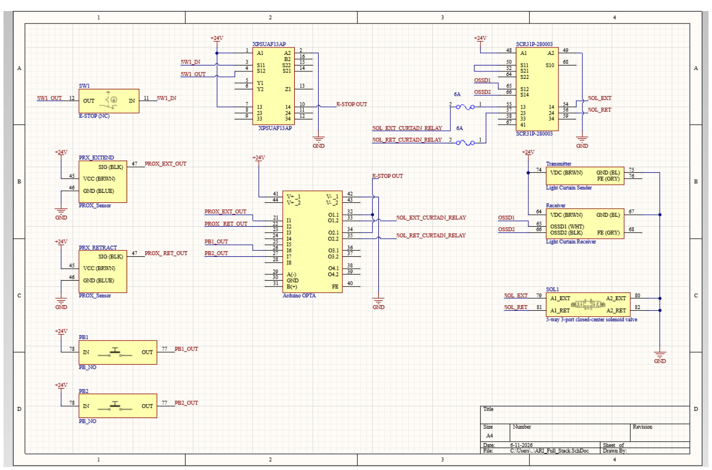
- Designed and built an automated electro-pneumatic press to insert 600,000 compression limiters, cutting cycle time 83% and eliminating operator injury and part-damage risk
- Engineered the safety architecture: anti-tie-down two-hand control, light curtain with safety relays, and a 5/3 closed-center valve for automatic retraction
- Designed the full electrical schematic in Altium and pneumatic circuit; released GD&T drawings backed by worst-case tolerance stack-ups
- Owned the full product launch: PFMEA, subsystem testing, process flow, work instructions, and SOPs



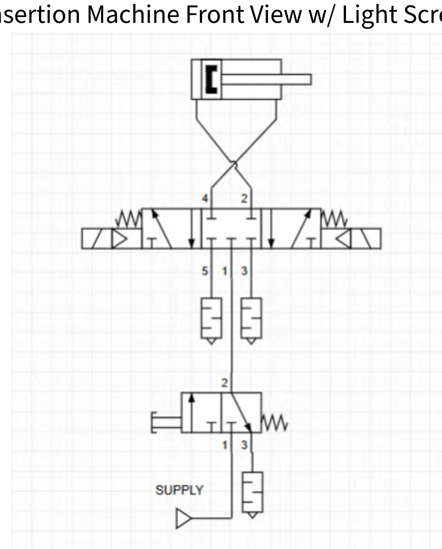
Insertion Machine Front View



Insertion Machine Front View w/ Light Screen



Electrical Schematic in Altium



Pneumatic Diagram

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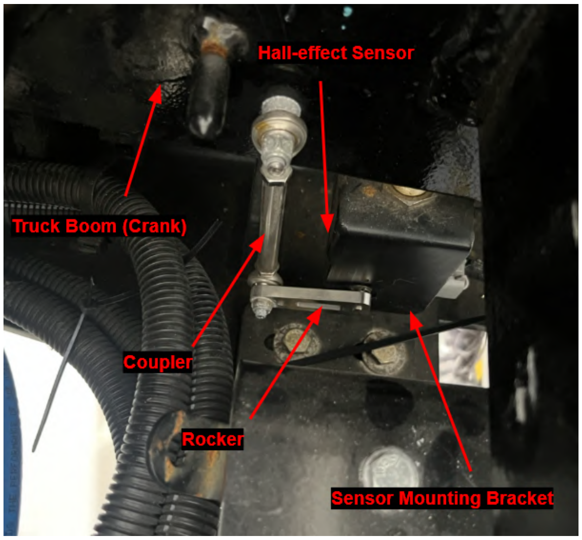
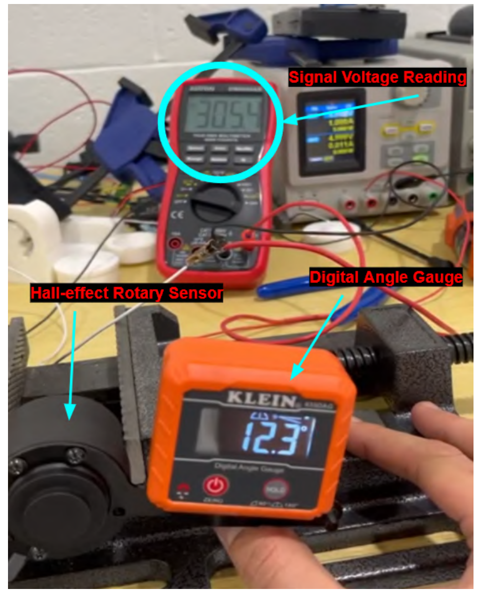
Electromechanical Truck Boom Angle Sensor @ Electrans Technologies

- Designed and built an IP67 boom-angle measurement system using a hall-effect rotary sensor driven by a four-bar linkage
- Synthesized the linkage for a 1:1 input-to-output angle ratio, preserving linearity across the full boom sweep
- Validated accuracy and repeatability to 0.01° by testing sensor signal voltage with a digital angle gauge



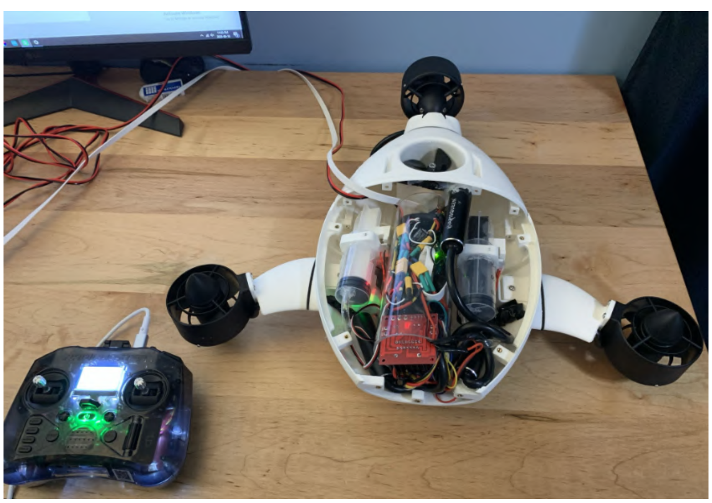
Assembled & Mounted System



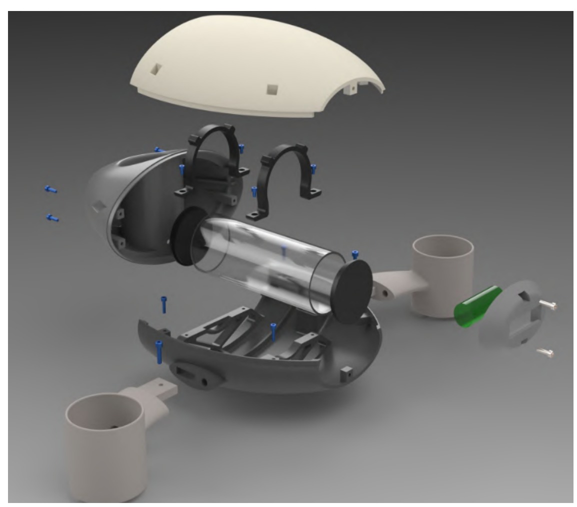
Sensor Validation

Amphibious Underwater Drone

- Designed an ROV via top-down master modeling and surface modeling in SolidWorks, driving parametric updates across a 15+ part assembly while holding G2 curvature continuity for better hydrodynamic efficiency
- Refined FDM parameters and applied acetone vapor smoothing to seal microporosity, achieving a watertight 550 g ABS frame at IPX7 with 15% neutral buoyancy margin
- Integrated a thrust-vectoring propulsion system and instrumented the ROV for submersion testing



Most Updated Harnessed Drone



Solidworks Render

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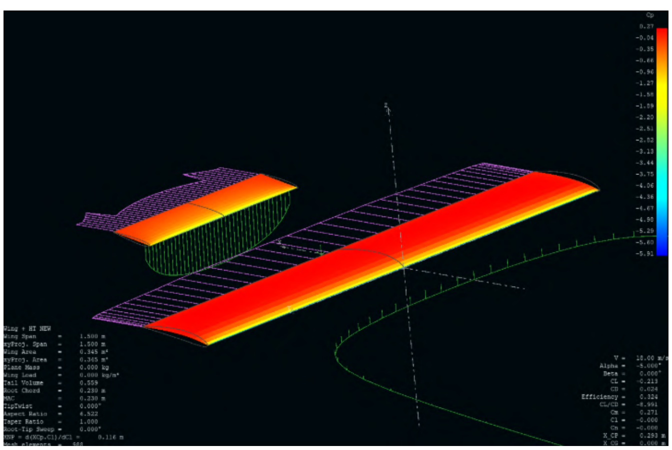
Fixed-wing UAV (Helios) & Competition Hex Drone (Valkyrie) @ WARG

- Led the full aerodynamic and mechanical design of the Helios STOL empennage, from configuration trade study through flight test; automated the stability-and-control design in a MATLAB/XFLR5 toolchain
- Actuated both elevator halves synchronously from a single servo via a coupled torque tube, saving weight and eliminating differential-deflection trim error; root-caused and redesigned in-service tail failures
- Managed mission-critical tasks on Valkyrie, including hardware mounting architecture for the fire-extinguishing task and Li-ion battery pack holders sized for flight loads and CG limits

Helios - Maiden Flight

Helios - Ground Testing



Helios - Aero Stability Simulation on XFLR5

Valkyrie - WARG Competition Drone

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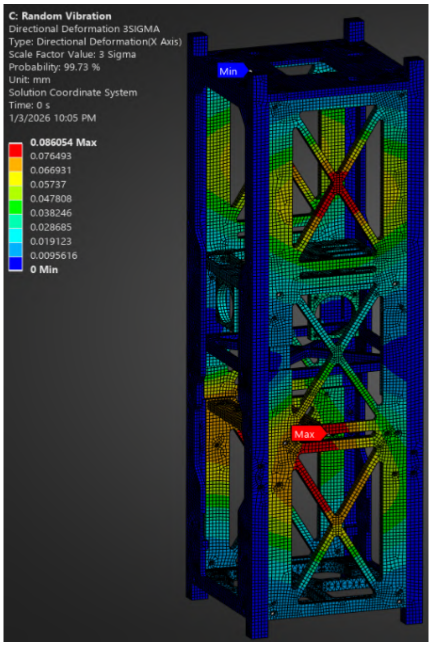
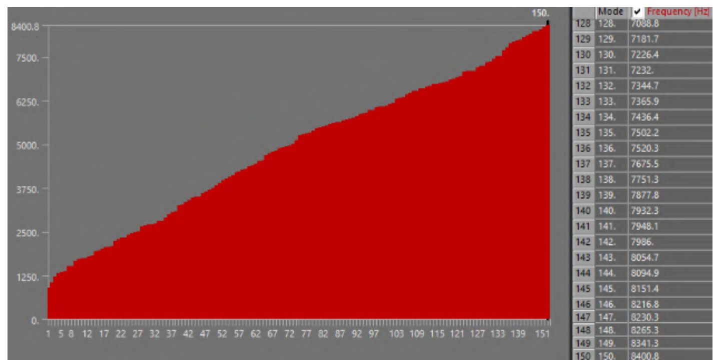
Non-Linear Quasi-Static, Modal, and Random Vibration FEA of CubeSat @ UW Orbital

- Solved the random vibration (PSD G) study to 1.5× PSD max frequency and ≥90% effective mass per axis, mapping 3-sigma displacement to find low-amplification IMU mounting zones
- Cut computation time 14 hours by partitioning geometry for ≥81% hex-dominant mesh with local refinement at stress gradients



Total Deformation (X Dir.) under Random Vibration

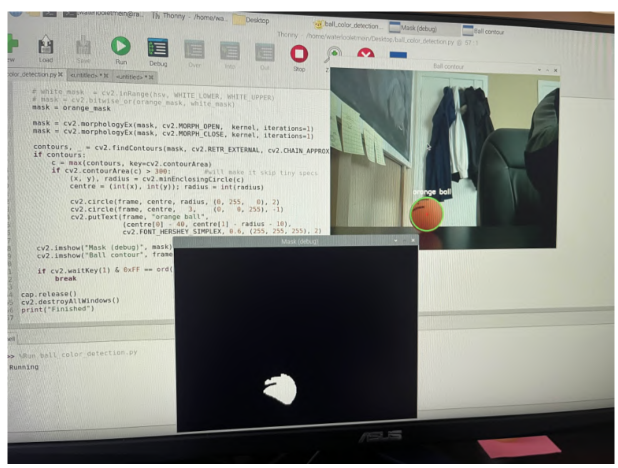


Chassis Resonant Frequency Modes

Autonomous Computer Vision Vehicle – C++, Python, Raspberry Pi, IR, ToF, ESP32

- Built a custom Python/OpenCV pipeline on a Raspberry Pi for ping-pong ball detection and range estimation, hitting 95% real-time detection and ±5 cm precision
- Developed C++ PID navigation on IR and ToF feedback, with a low-latency UART link between the Pi and ESP32 for real-time drive commands

Vehicle in Action



Python OpenCV Script Successful Detection of Ping Pong Ball

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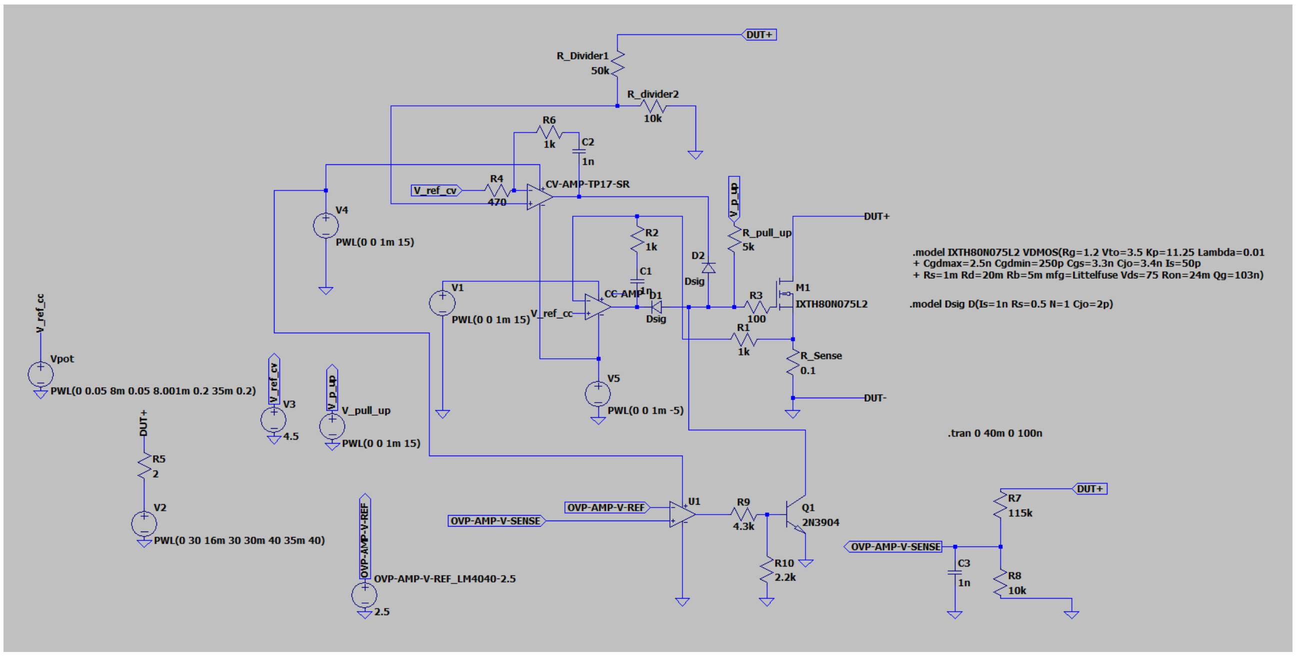
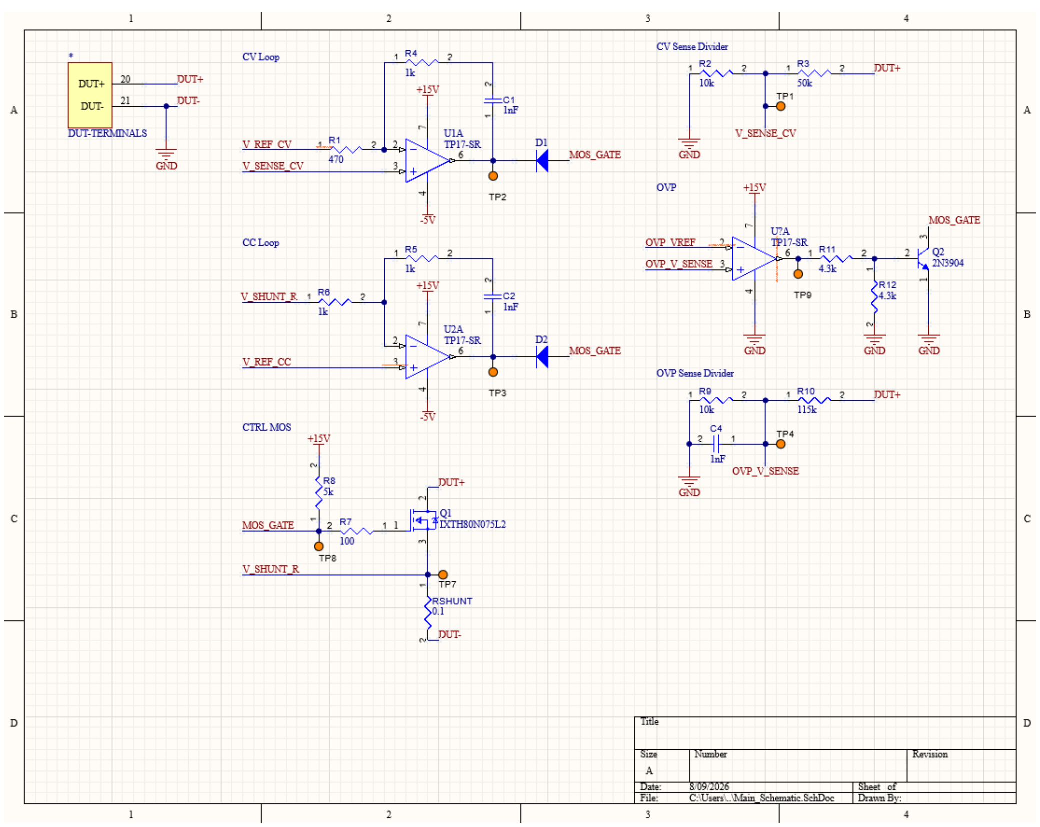
Programmable Electronic Load

- Designing a 50 W / 30 V / 5 A DC electronic load with a fully analog control loop, supporting constant voltage, constant current, and constant power modes; equipped with overvoltage and overcurrent protection
- Implemented mode arbitration via a diode-OR network of transconductance amplifiers, so the most restrictive loop takes control of the pass element without contention
- Ran transient analyses in LTspice to verify loop stability, mode-handoff behavior, and step response



LTSpice Transient Analysis Circuit



Altium Schematic & PCB Layout (In Progress)

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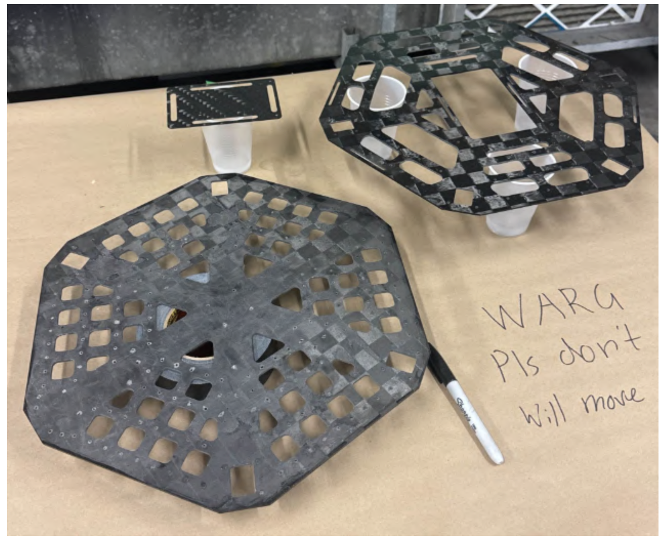
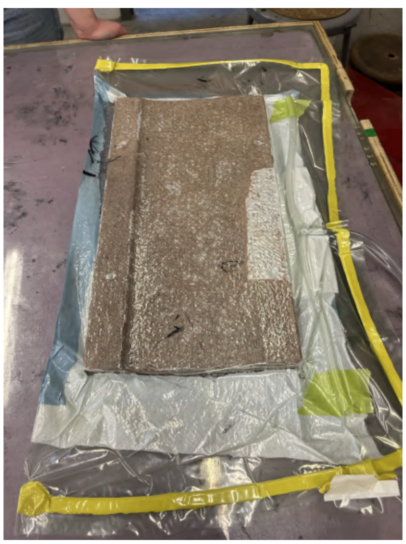
Composite Manufacturing @ WARG

- Fabricated high-strength composite UAV airframe components via split-mold tooling, wet layup, and foam-core layup
- Executed a 7-ply quasi-isotropic wet layup with manual consolidation and vacuum bagging; post-processed cured plates by sanding and epoxy-sealing edges to prevent delamination

Composite Wet Layup Process



Wet Layup Post-Processed



Split-Mold Vacuum Bagged

Split Molded Airfoil vs Foam Core Layup